Tutorial 8 Solution

Problem 1

In lecture 7, we learned about a result that if $f = h \circ g$, where h is increasing and convex, and g is convex, then f is also convex. We have also seen examples where the "outer function" h is convex but not increasing and therefore the composition fails to be convex.

However, if the "inner function" g is not just convex, but affine, then even if the "outer function" h is convex but not increasing, the composition $h \circ g$ is still convex.

(a) Prove the fact stated in the second paragraph; that is, if $g : S \to \mathbb{R}$ is affine and $h : \mathbb{R} \to \mathbb{R}$ is convex, where $S \subseteq \mathbb{R}^n$, then the composition $f \equiv h \circ g$ is convex. (Similarly, if g is affine and h is concave, then the composition $h \circ g$ is concave.)

(b) Use the fact in part (a) to show that

- $f_1(x, y) = (y - x)^2$ is convex; and
- $f_2(x, y) = -x^2 - 5y^2 - 4xy$ is concave.

(c) Find the global minima of $f_1(x, y)$.

(d) Find the global maxima of $f_2(x, y)$.

Solution:

(a) We need to show that $f = h \circ g$ is convex; that is, for any $\boldsymbol{x}_1, \boldsymbol{x}_2 \in S$ and every $\lambda \in [0, 1]$,

$$f\left(\lambda \boldsymbol{x}_1 + (1 - \lambda)\boldsymbol{x}_2\right) \leq \lambda f\left(\boldsymbol{x}_1\right) + (1 - \lambda) f\left(\boldsymbol{x}_2\right),$$

or equivalently,

$$h(g\left(\lambda \boldsymbol{x}_1 + (1 - \lambda)\boldsymbol{x}_2\right)) \leq \lambda h(g\left(\boldsymbol{x}_1\right)) + (1 - \lambda) h(g((\boldsymbol{x}_2)). \tag{1}$$

Since g is affine, it is both convex and concave, meaning that for any $\boldsymbol{x}_1, \boldsymbol{x}_2 \in S$ and every $\lambda \in [0, 1]$,

$$g\left(\lambda \boldsymbol{x}_1 + (1 - \lambda)\boldsymbol{x}_2\right) = \lambda g\left(\boldsymbol{x}_1\right) + (1 - \lambda) g\left(\boldsymbol{x}_2\right);$$

thus,

$$h(g\left(\lambda \boldsymbol{x}_1 + (1 - \lambda)\boldsymbol{x}_2\right)) = h(\lambda g\left(\boldsymbol{x}_1\right) + (1 - \lambda) g\left(\boldsymbol{x}_2\right)). \tag{2}$$

Since h is convex, for any $z_1, z_2 \in \mathbb{R}$ and every $\lambda \in [0, 1]$,

$$h\left(\lambda z_1 + (1 - \lambda) z_2\right) \leq \lambda h\left(z_1\right) + (1 - \lambda) h\left(z_2\right).$$

Taking $z_1 \equiv g(\boldsymbol{x}_1)$ and $z_2 \equiv g(\boldsymbol{x}_2)$, inequality (2) implies

$$h(g\left(\lambda \boldsymbol{x}_1 + (1 - \lambda)\boldsymbol{x}_2\right)) = h(\lambda g\left(\boldsymbol{x}_1\right) + (1 - \lambda) g\left(\boldsymbol{x}_2\right)) \leq \lambda h(g\left(\boldsymbol{x}_1\right)) + (1 - \lambda) h(g((\boldsymbol{x}_2)),$$

which yields the desired inequality (1). Thus, $f = h \circ g$ is a convex function.

(b) $f_1(x,y) = (y-x)^2$ is convex because $f_1 = h_1 \circ g_1$, where $h_1(z) = z^2$ is convex and $g_1(x,y) = x - y$ is linear (and therefore affine).

For f_2, note that

$$f_2(x,y) = -x^2 - 5y^2 - 4xy = -(x^2 + 4xy + 4y^2) - y^2 = -(x+2y)^2 - y^2.$$

By the fact we proved in part (a), we know that $g_2(x,y) := -(x+2y)^2$ is concave because $g_2 = u \circ v$, where $u(z) = -z^2$ is concave and $v(x,y) = x + 2y$ is linear (and therefore affine). Also, it is not difficult to see (for example, Tutorial 7 problem 2) that $h_2(x,y) = -y^2$ is concave. Then since $f_2(x,y) = g_2(x,y) + h_2(x,y)$, $f_2(x,y)$ is concave.

(c) From

$$\frac{\partial f_1(x,y)}{\partial x} = -2(y-x) = 0 \quad \Longleftrightarrow \quad y = x$$
$$\frac{\partial f_1(x,y)}{\partial y} = 2(y-x) = 0 \quad \Longleftrightarrow \quad y = x$$

we see that $(x,y) \in \mathbb{R}^2$ is a critical point of f_1 if and only if $y = x$. By part (b), f is convex, and therefore all the critical points, namely the points on the line determined by $y = x$, are global minima.

(d) From

$$\frac{\partial f_2(x,y)}{\partial x} = -2x - 4y = 0$$
$$\frac{\partial f_2(x,y)}{\partial y} = -10y - 4x = 0$$

we deduce that the only critical point of f is $(x^*, y^*) = (0,0)$. Since we determined in part (b) that f_2 is concave, $(x^*, y^*) = (0,0)$ is the unique global maximum.

Problem 2

Calculate the critical points of the function $f : C \to \mathbb{R}$ given by

$$f(x,y) = \ln(8y - x^2 + 4x),$$

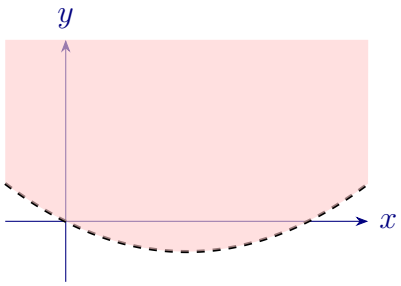
and find out all its local or global extrema (i.e., maxima or minima).

Solution: Since $h(z) = \ln z$ is a strictly increasing function, by a theorem in the lecture, the local and global maxima [minima] of $g : D \to \mathbb{R}$ given by $g(x,y) = 8y - x^2 + 4x$ coincide with the local and global maxima [minima] of $f = h \circ g$, respectively. From

$$\frac{\partial g(x,y)}{\partial x} = -2x + 4 = 0 \quad \Longleftrightarrow \quad x = 2$$
$$\frac{\partial g(x,y)}{\partial y} = 8 = 0 \quad \Longleftrightarrow \quad \text{impossible}$$

we conclude that no critical points exist.

Since C is an open set (see the graphical illustration below), every $(x,y) \in C$ that is not a critical point cannot be a local extremum, and hence cannot be a global extremum either. Thus, no local or global extremum exists.



Shaded region: C
(equivalent to $\{(x,y) \in \mathbb{R}^2 : y > \frac{(x-2)^2}{8} - \frac{1}{2}\}$)

Black dashed curve: boundary
($8y - x^2 + 4x = 0$, not contained in C)

Problem 3

Use extreme value theorem to show that the following maximization problems have a solution.

(a) $\max_{(x,y)\in C} f(x,y)$, where $f(x,y) = 2xy^2$, and

$$C = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

(b) $\max_{(x,y)\in D} g(x,y)$, where $g(x,y) = 4x + 7y$, and

$$D = \{(x,y) \in \mathbb{R}^2 : x + 2y \leq 3,\ x \geq 0,\ y \geq 0\}.$$

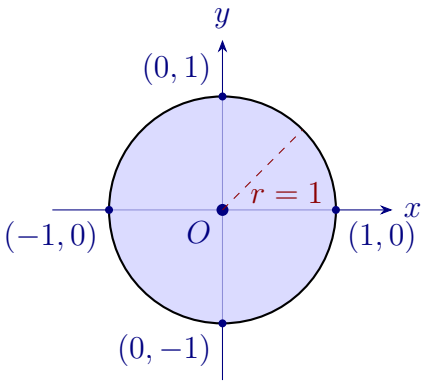
Solution:

(a) Observe that $f = 2(pr_1)(pr_2)(pr_2)$. Since the projections pr_1 and pr_2 are continuous, so is their product; moreover, multiplying a function by 2 doesn't change it continuity. Thus, the objective function f is continuous.

The set C is closed because it contains all its boundary points (every point in the set $\{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is a boundary point, which is contained in C), and it is also bounded because for all $(x,y) \in C$, $\|(x,y)\| \leq 1$. Therefore, C is compact.

Then by the extreme value theorem, the maximization problem $\max_{(x,y)\in C} f(x,y)$ has a solution.

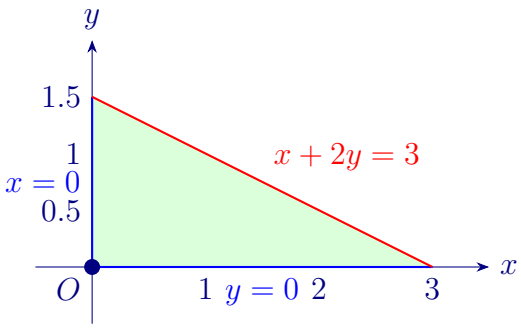
Graphically, C is the closed ball centered at $(0,0)$ with radius 1; its boundary is the circle in black.



(b) Observe that $g = 4pr_1 + 7pr_2$. Since the projections pr_1 and pr_2 are continuous, so are $4pr_1$ and $7pr_2$ (in the future, you can just say that any affine, and therefore linear, function is continuous). Then because the sum of continuous functions is continuous, the objective function g is continuous.

The set D is closed because it contains all its boundary points (every point in the set $\{(x, y) \in \mathbb{R}^2 : x + 2y = 3,\ x = 0,\ y = 0\}$ is a boundary point, which is contained in D), and it is also bounded because for all $(x, y) \in D$, $\|(x, y)\| \leq 3$. Thus, D is compact.

Graphically, D is the triangle area in the picture below. The set of boundary points is the lines in blue and red.



Problem 4 (Extra Practice Problem)

An artisan uses x_h units of labor and x_m units of a unique material to produce a product. Her profits if employing $(x_h, x_m) \geq 0$ units of the mentioned factors are

$$\pi(x_h, x_m) = 20x_h + 26x_m + 4x_h x_m - 4x_h^2 - 3x_m^2$$

Find the values of x_h and x_m that maximize the artisan's profits.

Solution: To find the global maxima of π, first show that the objective function π is concave. To that end, observe that if the term $4x_h x_m$ is absent, then $\pi(x_h, x_m)$ can be written as the sum of two functions, one is a function of x_h only, and the other is a function of x_m only, and both functions are concave. If this is the case, π must also be concave by the result in Tutorial 7 Problem 2. Therefore, we aim to "eliminate" the term $4x_h x_m$ by completing the square.

Note that $-4x_h^2 + 4x_h x_m - x_m^2 = -(2x_h - x_m)^2$, and therefore

$$\pi(x_h, x_m) = 20x_h + 26x_m - 2x_m^2 - (2x_h - x_m)^2.$$

Let $\pi_1(x_h, x_m) \equiv 20x_h + 26x_m - 2x_m^2$; by the result in Tutorial 7 Problem 2, it is a concave function. Also let $\pi_2(x_h, x_m) \equiv -(2x_h - x_m)^2$; by the result in Problem 1, π_2 is also a concave function. Then since $\pi = \pi_1 + \pi_2$, and the sum of concave functions is a concave function, π is also concave.

From

$$\frac{\partial \pi(x_h, x_m)}{\partial x_h} = 20 + 4x_m - 8x_h = 0 \iff 5 + x_m = 2x_h$$
$$\frac{\partial \pi(x_h, x_m)}{\partial x_m} = 26 + 4x_h - 6x_m = 0 \iff 3x_m - 13 = 2x_h$$

we see that the unique critical point of π is $(x_h^*, x_m^*) = (7, 9)$. Since π is concave, the unique critical point $(x_h^*, x_m^*) = (7, 9)$ must be a global maximum.

Summarizing, choosing $x_h = 7$ and $x_m = 9$ maximizes the artisan's profits.